Terminologies related to Data Science:-

① Data :

→ Raw facts and figures collected from various sources.
→ Raw facts and a piece of information.

Exp temperature readings, exam scores, customer feedback.

② ~~Definitions~~ - Dataset :-

(A) A structured collection of data, often in tabular form (rows and columns).

Exp- A CSV file of student names, marks and attendance. (B) unstructured data :- data without predefined format (emails, videos, images, social media)

③ Feature (Attribute/variable) :-

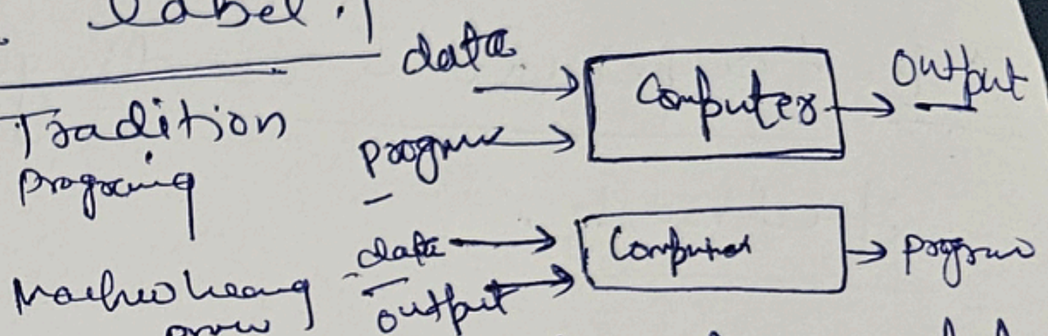
def :- An individual measurable (Quantifiable) property or characteristic of data. (calculable)

Exp In house price dataset — area, number of rooms, locations are features.

④ Label (Target variable)

Defⁿ:- The output or result you want to predict or classify.

Exp. whether a student passed or failed. (Yes/No) — this is a label.



⑤ Traing data:-

Definition:— A portion of the dataset used to train the machine learning model.

Exp. 80% of data used to teach a model. (how to classify spam emails.)

⑥ Testing Data:-

defⁿ.— The data used to test how well the trained model perfoms.

Exp. remaing 20% of data used to evaluate the model the spam classifier accuray.

⑦ model:-

defⁿ.— A mathematical or statistical formula that makes predictions or decisions using data.

exp A Linear regression model predicts house prices.

⑧ Algorithm:-

- defn: A set of rules or instructions the model follows to learn from the data.

exp: Decision tree, K-Nearest Neighbors, Linear regression.

⑨ Overfitting:-

defn:- The model learns the training data too well — including noise and outlier — and perform poorly on unseen data.

Exp. A model gets 100% accuracy on train data but only 60% on testing data.

⑩ Underfitting:-

The model is too (Highly) simple and cannot capture the underlying patterns of the data.

Exp A straight-line model used to predict complex curved patterns.

(11) Bias :-

defn error due to highly simplistic assumptions in the model.

High bias leads to underfitting.

(12) Variance :- error due to the model being too sensitive to small fluctuations in training data.

High variance :- leads to overfitting.

(13) Prediction :-

defn :- The output value given by a model for a new input data.

Exp: predicting tomorrow temperature..

(14) Accuracy :-

defn : A measure how many predictions made by the model are correct.

Exp: if model makes 90 correct predictions out of 100, accuracy is 90%.

(15) Confusion matrix :-

defn : A table used to evaluate the performance of a classification model.

Include: True positive, True negative, False positive, False negative.

Study hours	Output variable (Y)	(Ŷ)
2	0	0
4	1	0
5	1	1
1	0	0
6	1	1
3	0	1

Prediction \ Actual	1	0
1	TP 2	FP 1
0	FN 1	TN 2

(16) Precision, Recall, F-1 score

Precision:- of all positive predictions, how many were correct.

$$\frac{TP}{TP+FP}$$

Recall :- Of all actual positive, how many were Captured.

$$\frac{TP}{TP+FN}$$

F-1 Score: Harmonic mean of precision and recall.

$$\frac{2PR}{P+R}$$

(18) Exploratory Data Analysis (EDA):-

defn:- The process of visualizing and summarizing the data to discover patterns.

(19) Feature engineering:-

Creating new features or transforming existing one to improve model performance.

Exp Converting "Date of Birth" into "Age".

(20) Cross-validation:-

defn:- A technique to evaluate model performance by splitting data into multiple parts and testing on different subsets.

common type: K-fold cross validation.

Exp [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

K=5

$f_1 = [1,2]$
$f_2 = [3,4]$
$f_3 = [5,6]$
$f_4 = [7,8]$
$f_5 = [9,10]$

Round	Training	Testing (validation)
1	$f_2 f_3 f_4 f_5$	f_1
2		
3		
4		
5		

ias

In data sciem and machine learig, bias refer to the error introduced by making the model too simple to capture the real relationship of data.

- Simplistic Assumptions - what does it mean.
- The model assumes things that are not fully true about the data.

For exp. A linear model assumes the relationship b/w input and output is always a straight line
- which may not be true in real-world problems.

3. Fail to capture key patterns - what does this imply.
- When a model is too simple, it misses important trends, relationships or variations in the data.
- this leads to incorrect predictions bcz it ignores complexity in the real data.

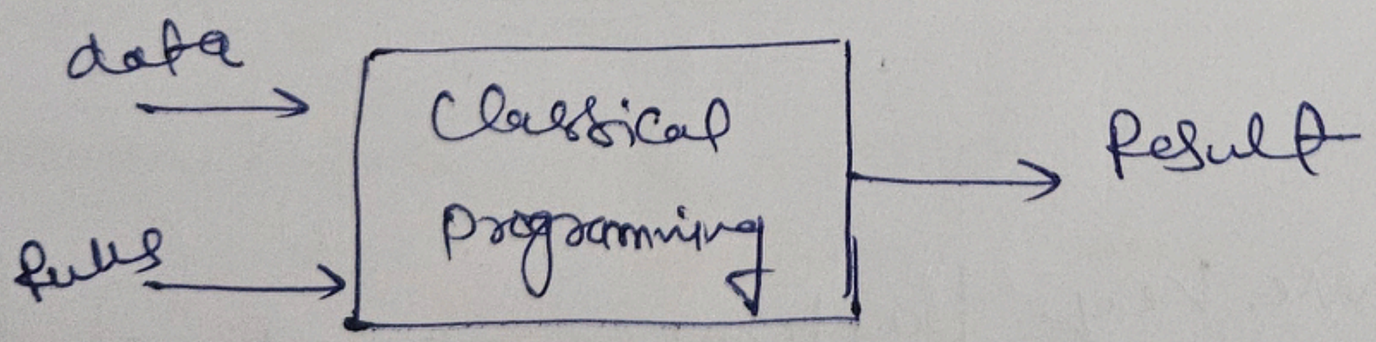
Varience :-

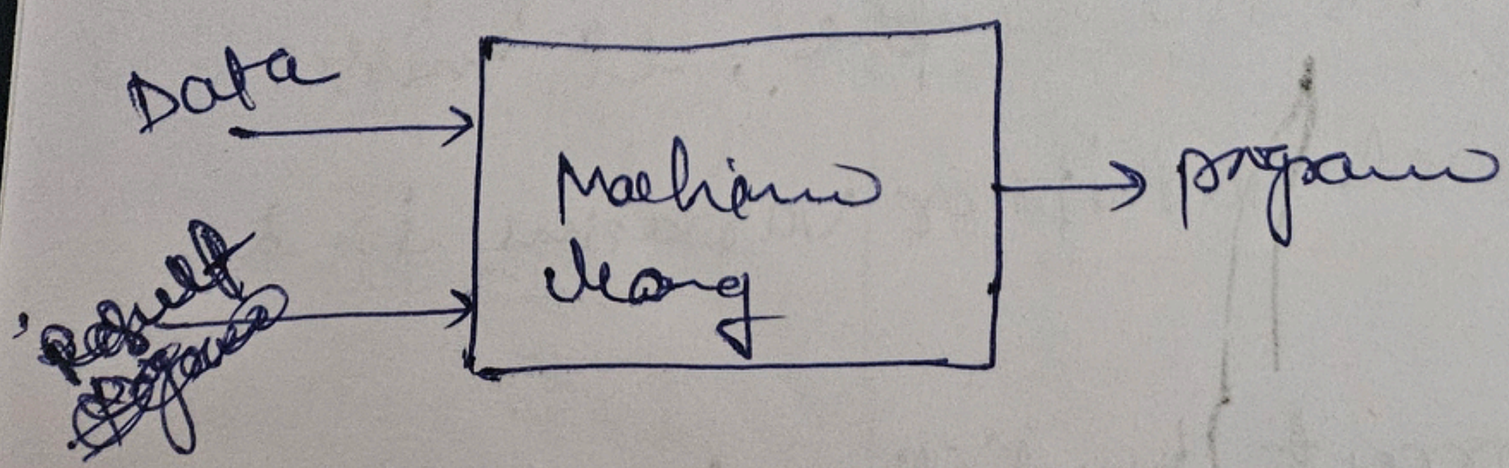
in data science and machine learig

Varience refers to the amount by which a model's predictions change when it's trained on different parts of the data.

② Sensitivity to small changes —.

- A High-varience model learns the traing data too well, including noise or random fluctuations
- if you slightly change the traing data, the model's predictions can change drastically.





Bias :- Bias is the error due to wrong assumptions made by model.

High bias = model is too simple and misses important patterns.

Goal :- predict marks scored based on study hours

- Model A : very simple (e.g. just predicts average marks)
- Model B : Balanced (good complexity)
- Model C : very complex (e.g. tries to perfectly fit every, data example.)

Model A says :

"No matter how much a student studies, it will just predict the average marks (say 60 marks)!

Even if a student studies for 10 hours, it still predict 60.

This is a high bias: the model ignores the real realtionship b/w study hours and marks.

Step (2) what is varience:-

Varience is the error due to too much sensitivity to taing data.

High varience :- model is too complex and gets confused by noise in data.

Exp

Model tries to perfectly mach every taing stelent data.

- 2 hours → 40 marks
- 3 hours → 55 marks
- 3.1 hours → 56 marks
- 3.2 hours → 58 marks

But when a new student comes with 3.15 hours to study it may give a wrong prediction.

this is high variance: model fits being data too well. but fails to generalize.

Step 3: Balanced model (Model B)

Model B learns that

more study hours generally lead to more marks. Let's draw a straight line like.

marks = 10*hours + 30

this model:-

- Captures the trend (low bias)

- Doesn't react to tiny change (low variance.

this is good. model.

Summary Table :-

Model	Bias	Varience	Description
Model A	High	Low	Too simple underfits.
Model B	low	low	Just right, generalize well.
Model C	low	high	Too complex, overfits

Exp

Imagine you are a teacher and want to build a model that predict if a student will pass or fail based on study hours.

Training.

You give the model a few examples (training data) where the input and output are known.

Study hours	Result (Pass / Fail.
2	Fail
4	Fail
6	Pass
8	Pass

The model learns from this exp.
it finds pattern like.
if Study hours ≥ 6, likely pass

Step 2

Now, you give a new data. to the model.
with telling the result like.

Study hours	Result (Hidden)
3	?
7	?

The model uses what it learned. to predict.

3 → Fail

7 → Pass

Note:-

- Baing is like studying with past question papers.
- Testing is like taking a final exam to check. how well you prepared, then you compare its predictions to the actual. results to see how well it performed.
- Baing:- To teach the model
- Testing:- to check if the model can make correct predictions on new/unseen data.